Week 17: NLP for social good

SCC453 Natural Language Processing and Language Models

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NLP Overview / Pipeline

Plan for today

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Definitions of NLP for social good

High level overview of some of the topics within this umbrella term, with example projects and links to specific NLP tasks

Guided literature survey task in small groups to focus on specific topics of your choice

(which will help you practice starting a literature review for NLP as part of the coursework ☺)

But first the bad news …

- Potential negative impacts of NLP, AI, DS, ML, DL …
  - Public Lecture: Does Artificial Intelligence pose an existential threat with Professor Bill Oxbury, Lancaster University https://www.youtube.com/watch?v=bQQzg9jcivM
- Bias in the data and predictions (see week 12's lecture)
- Ethical and legal concerns of data collection and exploitation (see week 12's lecture)
- Political bias and harmful recommendation systems in online social media
  - Instagram and self harm
  - Filter bubbles and echo chambers
  - The speed of the spread of fake news versus fact checked stories
- Vast quantities of power, cooling, CO2 required and produced by AI, LLMs, etc
  - See Lancaster's PARIS-DE project (https://wp.lancs.ac.uk/paris-de/), Widdicks et al (2024): https://doi.org/10.1145/3648234

- And potentially other negative impacts you can think of

Definition of NLP for social good

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- NLP for social good is an umbrella term that covers NLP applications which help the society by protecting it, by enhancing productivity, solving real world problems, or by helping those in need, including …
- NLP for enhancing work efficiency
- NLP for protecting society
- NLP in healthcare (including physical and mental health sectors)
- NLP for crisis situations
- NLP for accessibility (including NLP for language disabilities)
- Responsible and ethical approaches to doing NLP

1. NLP for enhancing work efficiency

- Spelling and grammar checkers, readability tools (word & character n-grams, soundex, edit distance)
- SMS predictive text (word & character n-grams)
- Search engines (PageRank algorithm and TF-IDF)
- Recommendation systems for online purchases, books, films, etc
- Speech recognition, synthesis, automatic captioning/signing and OCR for scanning and converting documents
- Machine translation (see SCC453 lecture week 20)
- Co-pilot for collaborative software code authoring (LLMs)
- Large Language Models generation of everything …

- Welcome to the smart assistant era (Gemini, Copilot in Excel, etc)
  - https://youtu.be/_fuimO6ErKI?si=lCiw63fbO2jv_NAQ
  - https://youtu.be/mNTGbi5ReMc?si=T8ipHThLNugqWMas
  - https://youtu.be/798VKff1v2U?si=d8bUGxxJFrtCa6yV
  - https://youtu.be/Ndj9Jk-tGKo?si=Ob5Bgx76YeL-3T7w
  - https://youtu.be/6kP_YlHEAhk?si=GmU4C45esBxg_aFA

Qualitative survey analysis: FreeTxt/TestunRhydd project (2022-3)

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- Surveys are widely used in many areas of professional practice, e.g. staff development, professional training, product design, testing as well as for many types of hotel, movie and product reviews
- Very little support for bilingual free-text survey and questionnaire data analysis in English and Welsh
- Follow on funding impact project building on CorCenCC project (National Corpus of Contemporary Welsh), we will develop an open access user friendly online interface
- Partners: National Trust Wales, Cadw and National Museum Wales
- https://ucrel.lancs.ac.uk/freetxt/

Further reading

- Brin, S. and Page, L. (1998) The anatomy of a large-scale hypertextual web search engine. http://infolab.stanford.edu/pub/papers/google.pdf
- Spärck Jones, K. (1972) A statistical interpretation of term specificity and its application in retrieval. Journal of Documentation 28, 11-21 https://www.staff.city.ac.uk/~sbrp622/idf.html
- Mitton, R. (1996) English spelling and the computer. Longman. https://titan.dcs.bbk.ac.uk/~roger/spellchecking.html
- Black, E., Garside, R. & Leech, G. (1993). Statistically-Driven Computer Grammars of English: The IBM/Lancaster Approach. Brill. https://brill.com/display/title/30130

2. NLP for protecting society … offensive/hate speech: motivation

- 5.24 billion people around the world now use social media (63.9%), 206 million new users have come online within the last year.
- The average daily time spent using social media is 2h 21m.
- 402M Tb of data generated online per day, Facebook is 5Pb per day; 500M posts on X/Twitter; 95M photos/videos on Instagram
- Content moderators: Meta has 15K, TikTok has 40K, X reduced by 20% since 2023 and dropped from 11 EU languages to 7
- Sources:
  - https://www.pewresearch.org/internet/fact-sheet/social-media/
  - https://datareportal.com/social-media-users
  - https://www.digitalsilk.com/digital-trends/how-much-data-is-generated-per-day/
  - https://www.thebureauinvestigates.com/stories/2024-05-17/explainer-what-is-content-moderation
  - https://www.cnbc.com/2024/05/08/eu-seeks-information-from-x-on-content-moderation-amid-dsa-probe.html

2. NLP for protecting society … offensive/hate speech: motivation

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Teenager Hannah Smith killed herself because of online bullying, says father

More than 30 pupils a day suspended for cyber bullying in UK schools

Hundreds of children were suspended or expelled from school for abuse on social media in just one year.

By MATTHEW DAVIS
22:25, Sat, Oct 15, 2022 | UPDATED: 22:25, Sat, Oct 15, 2022

One in 10 female gamers feel suicidal about the abuse they receive playing online

Missouri woman indicted in case involving MySpace-related suicide

By Linda Deutsch
AP Special Correspondent

LOS ANGELES – A Missouri woman was indicted Thursday for her alleged role in perpetrating a hoax on the online social network MySpace against a 13-year-old neighbor who committed suicide.

Lori Drew of suburban St. Louis, who allegedly helped create a MySpace account in the name of someone who didn't exist to convince Megan Meier she was chatting with a 16-year-old boy named Josh Evans, was charged with conspiracy and fraudulently gaining access to someone else's computer.

Megan hanged herself at home in October 2006, allegedly after receiving a dozen or more cruel messages, including one stating the world would be better off without her.

Salvador Hernandez, assistant agent in charge of the Los Angeles FBI office, called the case heart-rending.

"The Internet is a world unto itself. People must know how they can go before they n… They exploited a you… weaknesses," Hernandez … "Whether the defendant … have foreseen the results … sponsible for her actions."

Drew was charged w… count of conspiracy an… counts of accessing pr… computers without authorization to get information used to inflict emotional distress on the girl.

Drew has denied creating the … messages on MySpace.

believed media reports and public outrage helped move the case forward for prosecution.

"I'm thrilled that this woman … cumstances of their daughter's death.

Tina Meier has acknowledged Megan was too young to have …

Woman Indicted in MySpace Suicide Case
Under false persona, she leased girl who then killed herself

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Feds Seek 3 Years for 'Cyberbully' in Teen's Suicide
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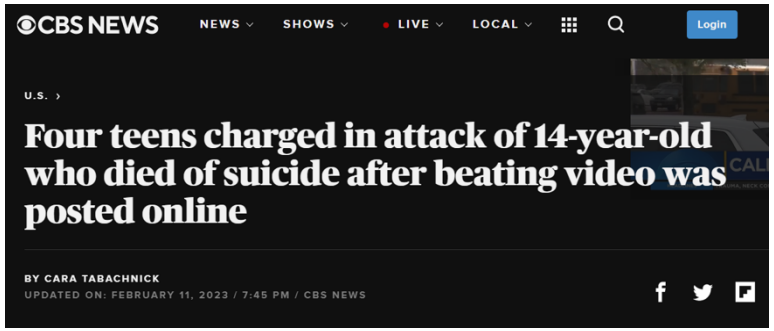
'Cyberbully' in Teen's Suicide
Probation is no deterrent: prosecutors

CYBER BULLYING AFFECTS REAL LIVES!



Texas child died by suicide during online game following cyberbullying, authorities said

Father of UK teenager blames social media for exposing her to 'bleakest of worlds'

2. NLP for protecting society … offensive/hate speech: shared tasks

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HASOC 2023

Hate Speech and offensive Content Identification in Multiple Languages

TRAC - 2022

Third (Virtual) Workshop on

Threat, Aggression and Cyberbullying

OffensEval

GermEval2021 - Toxic, Engaging, & Fact-Claiming Comments

Get the Data | Read the Proceedings | Contact the Organizers

2. NLP for protecting society … offensive/hate speech: datasets

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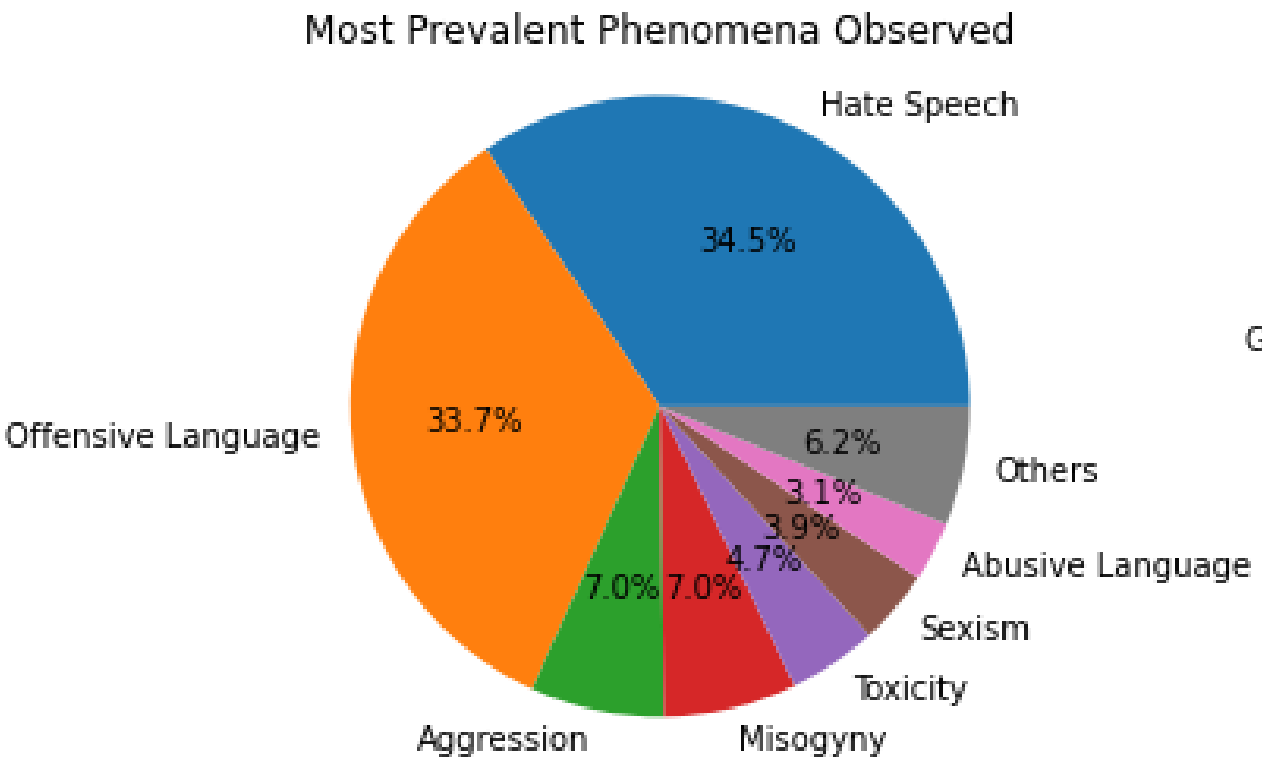
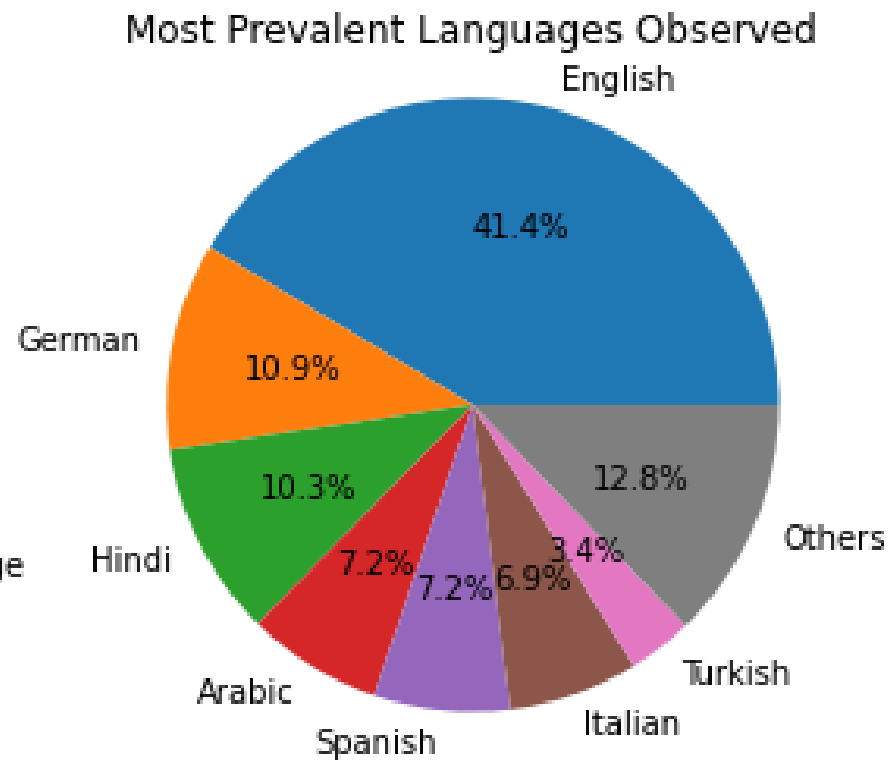
Name	Phenomenon	Source	Size	Reference
HatEval	Hate speech	T	13,000	[Basile *et al.*, 2019]
OLID	Offensive	T	14,200	[Zampieri *et al.*, 2019]
SOLID	Offensive	T	9,095,133	[Rosenthal *et al.*, 2021]
AMI	Misogyny	T	5,000	[Anzovino *et al.*, 2018]
EXIST	Sexism	T	7,646	[Rodríguez-Sánchez *et al.*, 2021 09]
TRAC-1	Aggression	F	12,916	[Kumar *et al.*, 2018]
TRAC-2	Aggression	F	5,000	[Kumar *et al.*, 2020]
HASOC 2019	Hate Speech	T	7,005	[Mandl *et al.*, 2019]
HASOC 2020	Hate Speech	T	4,521	[Mandl *et al.*, 2020]
HASOC 2021	Hate Speech	T	5,124	[Mandl *et al.*, 2021]
ALYT	Abusive	Y	19,915	[Bertaglia *et al.*, 2021]
AHSD	Hate Speech	T	24,802	[Davidson *et al.*, 2017]
Founta	Hate Speech	T	80,000	[Founta *et al.*, 2018]
Waseem	Hate Speech	T	16,914	[Waseem and Hovy, 2016]
Agrawal	Cyberbullying			[Agrawal and Awekar, 2018]
Gao	Hate Speech	Fox News	1,528	[Gao and Huang, 2017]
Gibert	Hate Speech	Stormfront		[de Gibert *et al.*, 2018]
Gab Hate Corpus	Hate Speech	G		[Kennedy *et al.*, 2022]
Toxic Spans Detection	Toxicity	Civil Comments	9,939	[Pavlopoulos *et al.*, 2021]
FRENK	Socially Unacceptable Discourse	F		[Ljubešić *et al.*, 2019]
HateXplain	Hate Speech	T,G	20,148	[Mathew *et al.*, 2021]

Table 1: English Offensive Language Datasets. F for Facebook, G for Gab, R for Reddit, T for Twitter, Y for Youtube.

2. NLP for protecting society … offensive/hate speech: datasets





Further reading

- European Union Digital Services Act: https://digital-strategy.ec.europa.eu/en/policies/digital-services-act-package
- UK online safety act: https://www.gov.uk/government/publications/online-safety-act-explainer/online-safety-act-explainer
- HASOC (2024) Hate Speech and Offensive Content Identification in English and Bangla https://hasocfire.github.io/hasoc/2024/
- OffensEval Shared Task: https://sites.google.com/site/offensevalsharedtask/home
- Workshop on Online Abuse and Harms (WOAH): https://www.workshopononlineabuse.com/
- Zampieri, M., Rosenthal, S., Nakov, P., Dmonte, A. and Ranasinghe, T., 2023. OffensEval 2023: Offensive language identification in the age of Large Language Models. Natural Language Engineering, 29(6), pp.1416-1435. http://doi.org/10.1017/S1351324923000517

3. NLP in healthcare: overview

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- A lot of research on electronic healthcare record analysis or mining of large scale collections of academic papers
- Ethics is particularly critical in such projects, plus explainability and false positives
  - E.g. Detecting depression and suicide risk
- Key challenges in extracting health and medicine terminology (see SNOMED and ICD taxonomies)

3. NLP in healthcare: recent Lancaster PhDs

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- Glorianna Jagfeld PhD: Talking about personal recovery in bipolar disorder: Integrating health research, natural language processing, and corpus linguistics to analyse peer online support forum posts
  - https://doi.org/10.17635/lancaster/thesis/2097
- Daisy Harvey PhD: Natural Language Processing Methods and Bipolar Disorder: Scoping Review
  - https://doi.org/10.2196/35928
- Lama Alsudias PhD: Using Arabic Twitter to support analysis of the spread of Infectious Diseases
  - https://doi.org/10.17635/lancaster/thesis/1648

3. NLP in healthcare: recent Lancaster projects

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- 4D PICTURE: will contribute to improved access to high-quality, sustainable health care, better health outcomes and more health equity (https://4dpicture.eu/)
- SAMS: three-year project investigating novel data and text mining techniques for detecting subtle signs of Cognitive Dysfunction in elderly computer users that could indicate the early stages of dementia (https://ucrel.lancs.ac.uk/sams/)
- Improving Peer Online Forums (IPOF): when people experience mental health difficulties, they often look for help online (https://www.lancaster.ac.uk/health-and-medicine/research/spectrum/research/ipof/)

Metaphor, cancer and end of life care (MELC)

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- Analysis of metaphorical language used to talk about cancer, dying and death: people 'fight' their cancer, 'win' or 'lose' their 'battle' against it, hope for a positive end to their cancer 'journey', and so on.
- 1.5M word corpus of interviews and online forum posts from patients, carers and healthcare professionals
- Methods: Manual analysis (MIP) and Wmatrix (Semantic analysis & concordancing)
- http://wp.lancs.ac.uk/melc/

G3 Warfare (e.g. *fight* as a verb, *battle*)
A1.1.1 General actions, making (e.g. *blast*, *confront*)
A1.1.2 Damaging and destroying (e.g. *destroy*, *shatter*)
E3– Violent/angry (e.g. *hit*, *attack*)
S8+ Helping (e.g. *defend*, *protect*)
S8– Hindering (e.g. *fight* as a noun)
X8+ Trying hard (e.g. *struggle*)

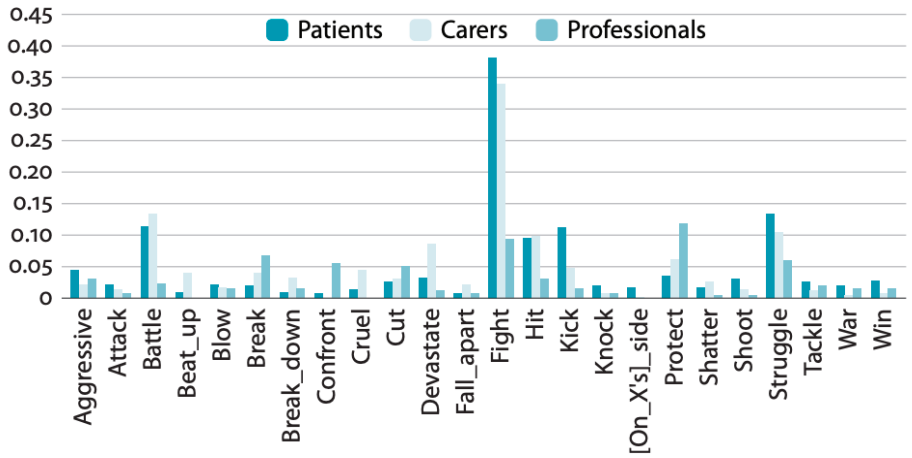

Figure 3. Relative use of most frequent Violence metaphors by each stakeholder group (per 1,000 tokens): Online forum posts

4D PICTURE

Design-based Data-Driven Decision-support (4D) Tools

Producing Improved Cancer Outcomes Through User-centered Research

The 4D PICTURE project aims to help cancer patients, their families, and healthcare providers better understand their options. It supports their treatment and care choices, at each stage of disease.

4D PICTURE

The project's primary objective is:

- To **improve decision-making** about cancer treatment by better predicting treatment outcomes by developing data-driven algorithms, resulting in decision-support tools for people with **breast cancer, prostate cancer, or melanoma**.
- To develop **conversation tools** for cancer patients, their significant others, clinicians, and citizens based on text mining analyses of patient experience 'big' data and citizen science methods.

Further reading

- Healthcare text analytics conference series: https://healtac2026.github.io/
- HealTAC2024 two keynotes on LLMs in healthcare and MIMIC dataset creation https://www.youtube.com/playlist?list=PLO93yuPJT61-BFBDGef8ZvzKwTx-S2z9c
- Biomedical Natural Language Processing Workshop (BioNLP) workshop series https://aclanthology.org/venues/bionlp/
- Workshop series on Computational Linguistics and Clinical Psychology https://clpsych.org/
- SNOMED and ICD taxonomies:
  - https://www.snomed.org/
  - https://digital.nhs.uk/services/terminology-and-classifications/snomed-ct
  - https://icd.who.int/en

Yet further examples of NLP for social good

- Age/gender prediction, language change over time
  - Cyber security
  - Isis project: Online Child Protection
  - Extremism and counter extremism
  - Dementia detection
- Text simplification and MT
  - Tools to help improve accessibility and empower people with disabilities
  - Crisis situations e.g. war or natural disasters

Further reading

- Edwards, M., Rashid, A., & Rayson, P. (2015). A systematic survey of online data mining technology intended for law enforcement. ACM Computing Surveys, 48(1). https://doi.org/10.1145/2811403
- May-Chahal, C., Mason, C., Rashid, A., Walkerdine, J., Rayson, P., & Greenwood, P. (2014). Safeguarding cyborg childhoods: incorporating the on/offline behaviour of children into everyday social work practices. British Journal of Social Work, 44(3), 596-614. https://doi.org/10.1093/bjsw/bcs121
- Rashid, A., Baron, A., Rayson, P., May-Chahal, C., Greenwood, P., & Walkerdine, J. (2013). Who am I? Analysing Digital Personas in Cybercrime Investigations. Computer, 46(4), 54-61. https://doi.org/10.1109/MC.2013.68
- Workshop on Tools and Resources for REAding DIfficulties (READI) https://aclanthology.org/venues/readi/
- NLP4Disability: Workshop on Natural Language Processing (NLP) for Disability https://nlp4disability.github.io/

Any questions so far?

Practical task: mini literature review

Mini literature review: instructions

- Form small groups (1-4 people)
- Pick a sub-topic of NLP for social good (it can be one I haven’t mentioned so far)
- Spend 30-40 minutes researching this task
- Record your results in a shared document or spreadsheet
- Report back: what topics have you found? Any unexpected results? Reflect on the method

Mini literature review: instructions

- Target: Pick a potential topic or search for a higher level topic first
- Method: Think about your starting points, which journals or venues and search terms are useful, how far to follow the citation networks?
  - https://aclanthology.org/
  - https://dl.acm.org/journal/csur
  - https://scholar.google.com/
  - https://ieeexplore.ieee.org/Xplore/home.jsp

These will also be good places to start when researching your background section in the coursework report

Mini literature review: other places to start

- Workshop series
  - Symposium on NLP for Social Good (NSG) https://nlp4social.github.io/NSG/
  - Workshop on NLP for Positive Impact https://sites.google.com/view/nlp4positiveimpact/
  - Workshop on Gender Bias in Natural Language Processing https://gebnlp-workshop.github.io/
  - Clinical Natural Language Processing https://clinical-nlp.github.io/
- NLP for social good initiative: https://nlp4sg.vercel.app/
- Semantic Evaluation Shared Tasks: https://semeval.github.io/
- Karamolegkou et al. (2026) NLP for Social Good: A Survey and Outlook of Challenges, Opportunities, and Responsible Deployment. To be presented at EACL2026. https://doi.org/10.48550/arXiv.2505.22327

Any final questions?

Week 17 Lab Tasks

1. Use the time in the lab this week to continue your investigations into NLP for social good (from the lecture) … or
2. Begin (or continue) your literature review, and other work towards, your coursework report